Getting to Know HAPS

"Cell Towers in the Sky" That Could Change Life in Thailand

HAPS (High Altitude Platform Stations) are solar-powered, uncrewed aircraft that float almost motionless in the sky about 20 kilometres up — roughly twice as high as a passenger jet, above the clouds and all the weather, yet hundreds of times lower than a satellite. In simple terms, think of them as "mobile-phone towers that hover in the sky." And because they run on sunlight, they can stay up there for months at a time without ever coming down to refuel.

Because they sit so high and stay over the same area continuously, a single HAPS can cover a very large region — like lifting one tower into the sky that does the job of dozens of towers on the ground. Just as importantly, the signal is fast with almost no delay, unlike a connection routed through a satellite that is far, far away.

What Can HAPS Do for Us?

1) Bring internet and mobile coverage to remote areas — Mountain villages, islands out at sea, and border regions that today have weak or no signal would be able to make calls, receive calls, and use the internet on their existing phones, with no special equipment required.

2) Help during disasters — When a major flood or earthquake strikes, ground-based towers are often knocked out or damaged — cutting off communication at the very moment people need it most. A HAPS can be sent up quickly to restore the signal over a disaster zone, so that calls for help, search-and-rescue, and coordination can happen without delay.

3) Watch over farmland and warn of danger early — With high-resolution cameras and sensors, a HAPS can keep an eye on fields around the clock. It can tell which areas are about to flood, which are facing drought, and where crops are showing signs of disease or pests — so farmers can prepare in time, reduce losses, and use water, fertiliser, and pesticide more efficiently.

4) Strengthen national security — A HAPS acts like a "watchtower in the sky," keeping continuous watch over borders and territorial waters. It also serves as a communications network the country can control on its own, without depending entirely on foreign systems.

Why This Matters for Thailand

Thailand has mountains, islands, and border regions that signals struggle to reach. It is home to around **12 million farmers** and faces floods and droughts almost every year. The floods in late 2024 alone caused tens of billions of baht in damage to the farming sector. HAPS is a tool that can address all of these challenges at once — communications, agriculture, disaster response, and security.

Right now, many countries such as Japan and the United Kingdom, along with global manufacturers like Airbus and SoftBank, are racing to develop HAPS and prepare to launch real services. If Thailand develops this technology for itself, the benefits would be threefold: closing the technology gap for people in remote areas, creating new jobs and industries at home, and achieving technological self-reliance instead of depending entirely on foreign systems.

In Short

- **HAPS = a cell tower in the sky** that floats 20 km up on solar power and stays aloft for months.
- **Four key benefits:** internet/mobile coverage in remote areas, signal recovery during disasters, support for agriculture, and national security.
- **A great fit for Thailand** because we have many remote areas, a large farming population, and frequent disasters.
- **Worth developing ourselves** to close the access gap, create local jobs, and build technological self-reliance.

Note: This is a condensed, plain-language edition based on the full HAPS White Paper, which draws on information from the ITU, the HAPS Alliance, Airbus, SoftBank, and other relevant organisations.